Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/15/2025
PI Name:	Matson, Douglas Ph.D.		
Project Title:	PSI Revisiting EML Video Images from EML Work (PREVIEW)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-PSI E.8 Physical Sciences Informatics
Start Date:	12/15/2023	**End Date:**	12/14/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nawer, Jannatun Ph.D. (Tufts University)		
Grant/Contract No.:	80NSSC24K0468		
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Task Description:	Abstract: The aim of this study is to leverage the existing dataset from the NASA Physical Science Informatics (PSI) database to re-evaluate thermophysical property data from the TEMPUS EML (Electromagnetic Levitation) on board the Space Shuttle Columbia. The success of evaluation of surface tension, by measuring the frequency response of an oscillating droplet, and viscosity, by evaluating how the oscillations dampen out, greatly depends on the quality of the acquired dataset and thoroughness of analysis methodology. Maintaining high accuracy is critical as it directly influences subsequent data analysis and interpretation tasks. The approach selected involves developing an automated video analysis protocol due to the condition of the archived data. This project conducted preliminary studies on the quality of the dataset and subsequent analysis of surface oscillations based on videos of STS-65 and STS-94. Video data were successfully acquired from PSI database and PI tapes which were evaluated for surface oscillation patterns. This study takes an extensive deep dive into the image analysis methodology using Optical Character Recognition and sub pixel edge detection. Aims and Objectives: The key scientific questions that were initially proposed to address during this study were: 1. Re-evaluate old data based on insights gained during subsequent years of investigation. 2. Study and report accuracy and precision during thermophysical property measurement comparing previous microgravity experiments to those currently ongoing on the ISS. Based on the data obtained the scope of this project was re-evaluated and the following questions were addressed: 1. Evaluate surface oscillations from shuttle data videos obtained from PSI database and PI archive 2. Re-evaluate surface tension based on new information on impact of sample deformation. 3. Conduct a detailed uncertainty analysis on the analysis methodology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If the aims of the proposal are achieved: If the aims of the proposal are achieved: (i) Advance the understanding of how the stored data from shuttle area data is influenced by previously unknown measurement biases. In the future this will help the researchers to conduct better experiments to leverage the unique environment of microgravity. (ii) Initiate activity across the property measurement community to encourage standards reporting of uncertainty of measurements to help the scientific and industrial design communities to better understand error propagation and to increase the fidelity of risk assessment for design of aerospace hardware supporting human exploration of space.
Task Progress:	A summary of the key findings from the project: • Video data were successfully acquired from the NASA Physical Sciences Informatics (PSI) database and Principal Investigator (PI) tapes. • De-interlacing was successfully performed for both video types. • Timecode digitization detection was conducted using OCR with an accuracy of 98%. • Timestamps within video frames show incorrect order of interlacing in both videos along with missing data points. • Surface oscillation data was successfully obtained from the PI VHS tapes. Frequency identification method needs to be further inspected. • Several sources of uncertainties were identified for the reported methodologies.
Bibliography Type:	Description: (Last Updated: 05/05/2026)